



CHAPTER III

METHODOLOGY

This chapter discussed the tools and methods used to develop and evaluate the proposed system in a systematic manner. This study utilized the Systems Development Life Cycle (SDLC) as the primary approach for the planning, analysis, design, development, testing, and implementation of the system. The chapter presented the operational framework of the project and included important system modeling tools such as Data Flow Diagrams (DFD) with context and multi-level explosions to illustrate how data flowed within the system, Entity Relationship Diagram (ERD) to define the data structure and relationships of the database, and Unified Modeling Language (UML) Use Case Diagrams to show how users interact with the system and its functionalities.

A Data Dictionary was also included to clearly define the data elements, fields, and terminologies used in the system. This helped ensure consistency, clarity, and proper understanding of all data used in the proposed system. The methodology further described the logical process and technical features of the Automotive Service Operations and Decision Support System for Multi-Branch Service Centers, including centralized branch management, automotive service operations, customer and vehicle profile management, quotation and job order processing, vehicle service status monitoring, inventory stock tracking, forecasting, and decision support system (DSS) functionalities.

In addition, this chapter identified the hardware and software requirements necessary for system development and deployment, presented the project timeline, and discussed the testing procedures such as functionality testing, user acceptance testing, and usability evaluation to ensure that the system was accurate, reliable, and easy to use. The chapter concluded by identifying the respondents involved in the study and addressing ethical considerations to ensure data privacy, security, and integrity throughout the research process.



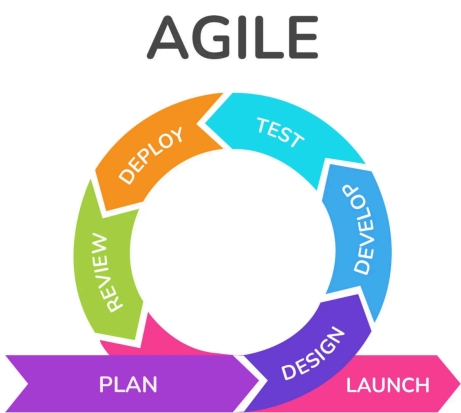
3.1 Systems Development Life Cycle

The Systems Development Life Cycle referred to the framework used to organize, plan, and manage the process of creating information systems. In this research, a mixed-methods approach combining descriptive and developmental research was employed. The descriptive component analyzed the existing automotive service operations in terms of system capability, service efficiency, and inventory management practices. It also evaluated the current workflow of the multi-branch service centers using ISO/IEC 25010:2011 for system quality evaluation, particularly in terms of functionality, reliability, usability, and efficiency, and ISO 9241-11:2018 for usability evaluation to ensure that the system would be easy to use and user-centered. Test cases were also used to validate system functionality and performance.

Developmental research was utilized during the planning, design, implementation, and testing phases, incorporating appropriate techniques and tools to improve the overall service operations and system performance. The system was developed using the Agile approach within the Systems Development Life Cycle (SDLC) model for the project. This approach allowed the system to be developed in iterative cycles, enabling continuous improvement, flexibility, and faster adaptation to changes during development. Each iteration included planning, development, testing, and evaluation to ensure that the system met the required standards and user needs. This method helped ensure that the proposed system was efficient, reliable, user-friendly, and capable of supporting centralized operations, inventory forecasting, and decision support across multiple service center branches.

Figure 1

Agile Development Model





As shown in the previous page, Figure 1 shows the Agile Systems Development Life Cycle used in the study. The system development life cycle guided proponents in developing the Automotive Service Operations and Decision Support System for Multi-Branch Service Centers. This software development model involved iterative phases such as planning, requirement analysis, system design, development, testing, and evaluation conducted in repeated cycles or sprints. The method was suitable for developing the system due to its flexible and adaptive nature, allowing continuous improvements throughout the development process. Each iteration produced a functional component that could be tested and evaluated to ensure quality and performance. Pressman and Maxim (2020) supported this approach, emphasizing that Agile-based SDLC models enhance system adaptability, continuous delivery, and user-centered development in modern information systems. Similarly, the proposed system incorporated centralized service management, real-time inventory tracking, forecasting, and decision support modules to improve efficiency and operational control across multiple branches.

Phase 1 – Requirements Planning Phase

During this phase, the researchers prepared the foundation of the study by conducting brainstorming and gathering necessary information related to the system. Initial research and review of related studies were performed to understand existing automotive service operations and identify gaps in the current process. The proponents also consulted with the adviser to refine the project title and finalize the study direction. After analyzing the gathered data, the system's requirements, scope, and objectives were clearly defined. This phase lasted approximately 20 days, as reflected in the Gantt Chart.

Phase 2 – User Design Phase

During this phase, the researchers focused on designing the structure and overall flow of the system. Chapter 1 to Chapter 3 were prepared alongside the system design process. The proponents created system diagrams such as context diagram, data flow diagrams, entity relationship diagram, and use case diagram to visualize how the system would function. In addition, system design planning, user interface layout, and database schema were developed to ensure that the system would meet user requirements. This phase also included revisions based on feedback from the adviser. This phase lasted approximately 40 days, as shown in the Gantt Chart.



Phase 3 – Rapid Construction Phase

At this stage, the system was developed based on the approved design. The researchers performed prototype development, coding of system modules, and integration of system features such as customer management, service operations, inventory tracking, forecasting, and decision support functionalities. Testing was conducted during development to ensure proper functionality of each module. Test cases were used to validate system performance, while ISO/IEC 25010:2011 was applied for system quality evaluation and ISO 9241-11:2018 for usability evaluation from the users' perspective. Debugging and system refinements were continuously performed to address issues and improve performance. This phase lasted approximately 130 days, as reflected in the Gantt Chart.

Phase 4 – Cutover Phase

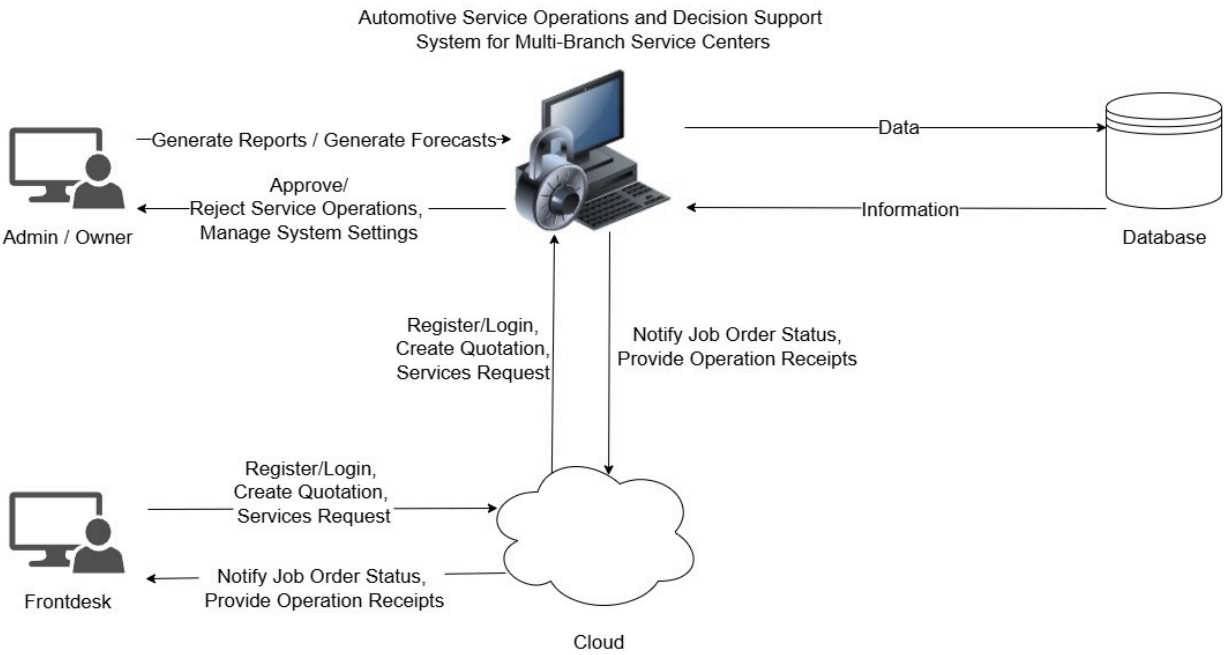
During the Cutover phase, the system was prepared for final deployment and presentation. The researchers conducted final revisions, documentation, and mock defense preparation to ensure readiness for evaluation. The system was finalized, and all necessary documents were completed, including formatting, printing, and binding. After completion, the final documentation was submitted. This phase also included preparation for presentation and final system validation. This phase lasted approximately 30 days, as shown in the Gantt Chart.



3.2 Operational Framework

Figure 2

Automotive Service Operations and Decision Support System for Multi-Branch Service Centers



3.3 Data Flow Diagram

Figure 3

Context Diagram (Level 0) of Automotive Service Operations and Decision Support System for Multi-Branch Service Centers

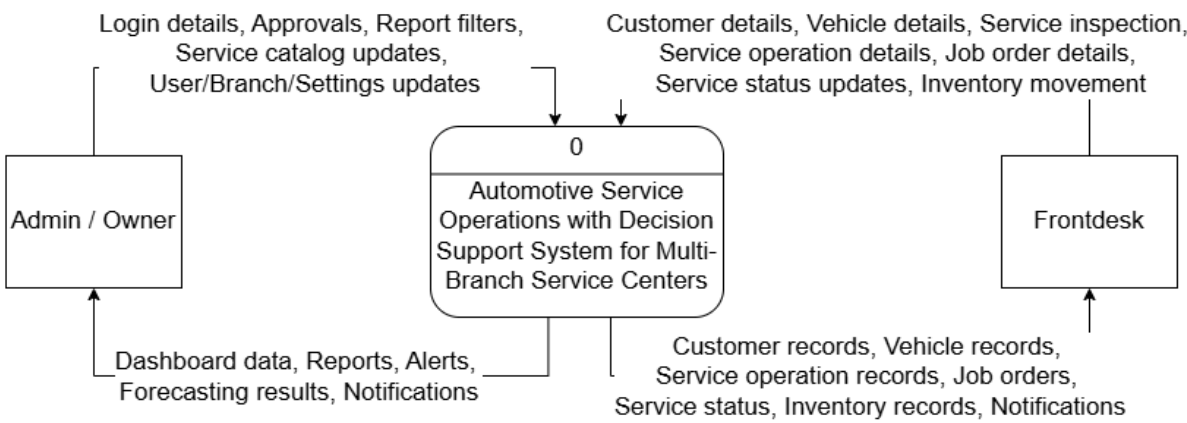
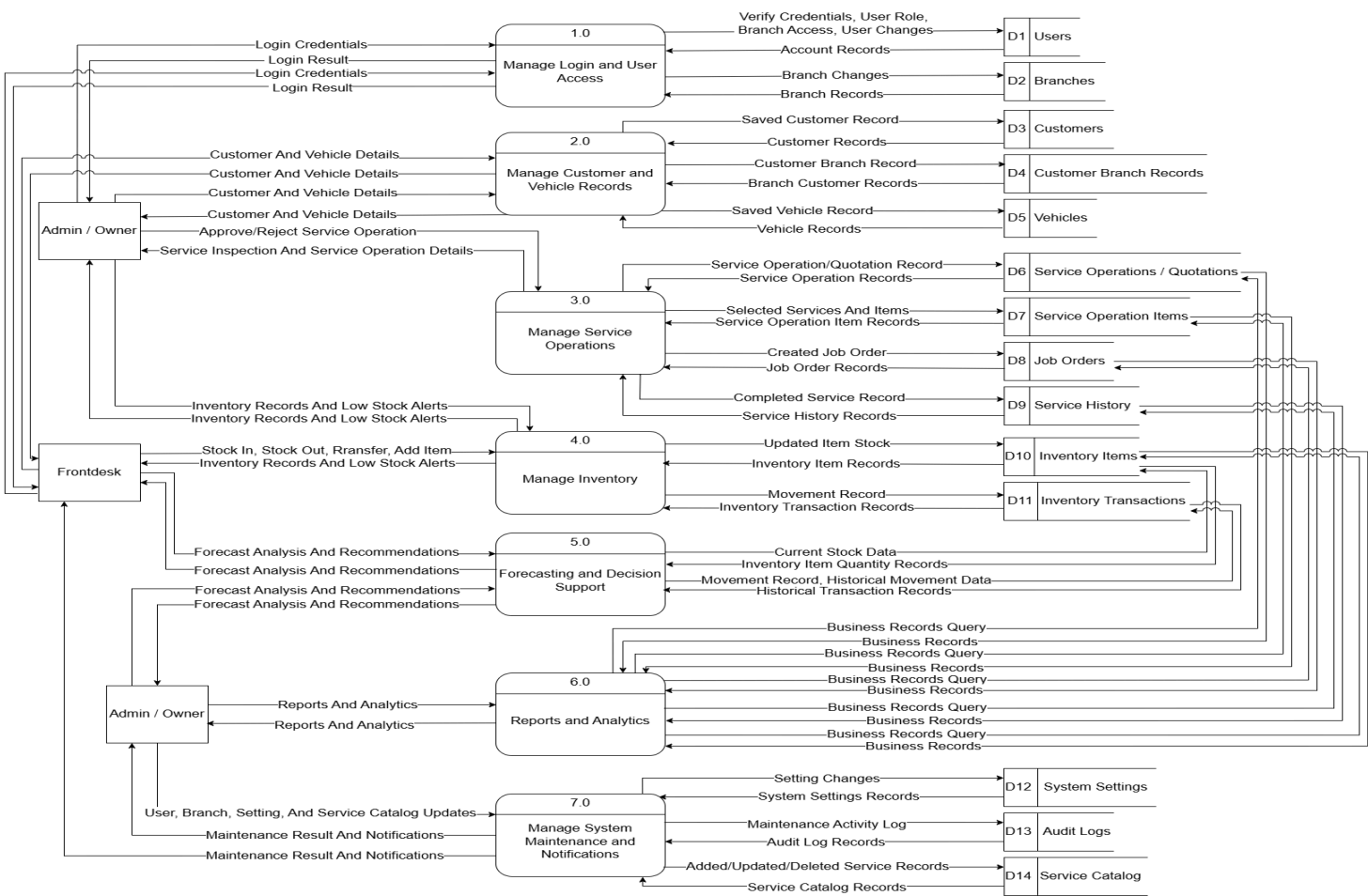




Figure 4 shows the context diagram of the developed system, illustrating the primary data and information flows between the system, the Front Desk, and the Admin/Owner. The inputs from the Front Desk include customer and vehicle details, service inspection records, service operation updates, job order information, service status updates, payment details, and inventory or stock movement data. The inputs from the Admin/Owner include user and branch management data, service maintenance requests, stock updates, transfer approvals, report filters, forecasting requests, and service catalog updates. The outputs generated by the system include updated customer and vehicle profiles, service operation records, quotation forms, job order summaries, updated service and payment statuses, stock availability reports, inventory movement records, low-stock alerts, forecast summaries, replenishment recommendations, consolidated branch performance reports, and system notifications. The diagram highlights the system's role as a central processing unit, coordinating inputs and outputs between users and managing data integrity, decision support, and operational reporting for multi-branch service centers.

Figure 4

Data Flow Diagram Level 1 Explosion of Automotive Service Operations and Decision Support System for Multi-Branch Service Centers

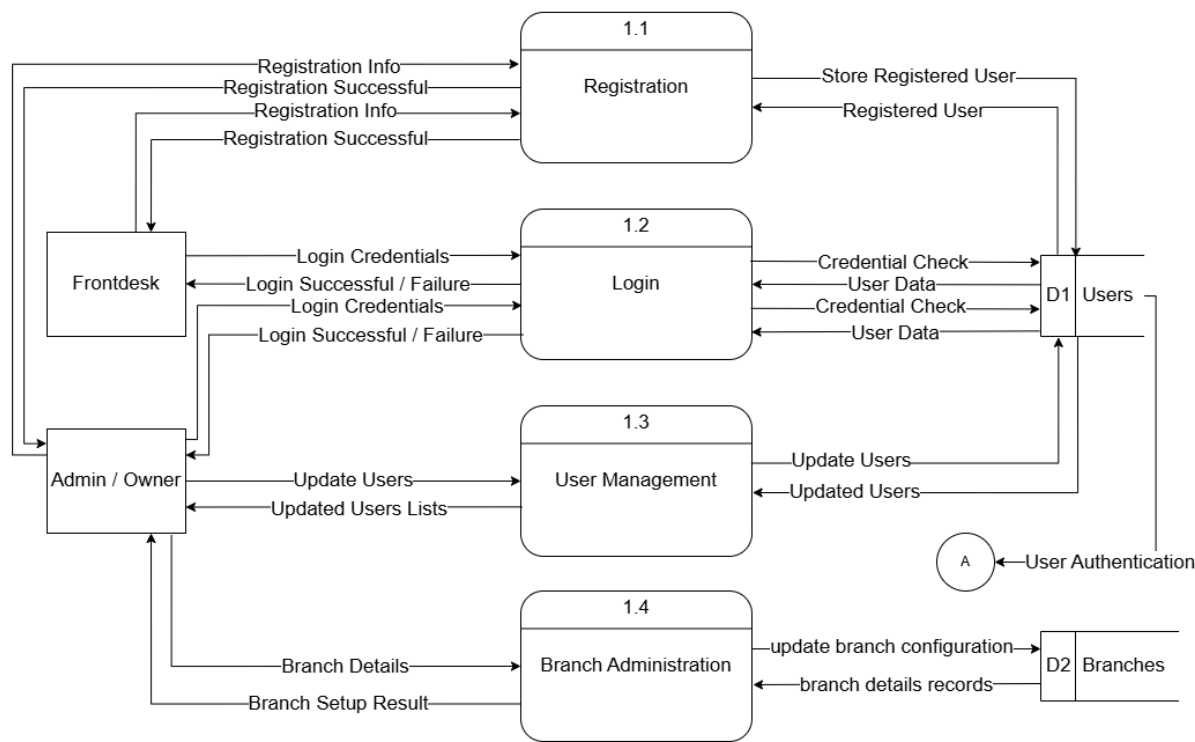




The Level 1 Data Flow Diagram illustrates how the system operates as a central hub connecting the Admin/Owner and Front Desk with the core processes of the automotive service operations and decision support system for multi-branch service centers. Users log in and the system verifies credentials, roles, and branch access while managing user accounts and branch records. Customer and vehicle information is entered and maintained, including branch relationships and service operation details, while service operations, job orders, and completed service records are managed and stored for reference. Inventory movements such as stock-in, stock-out, transfers, and low-stock alerts are recorded, and the system provides forecasting and decision support by analyzing current and historical stock data to recommend restocking and transfers. Reports and analytics consolidate business records to provide insights, dashboards, alerts, and performance summaries for decision-making. The system also manages maintenance activities, updates, and notifications, ensuring that all data flows seamlessly, remains accurate, and supports the operations of multiple branches effectively.

Figure 5

1.0 Manage Login and User Access Explosion



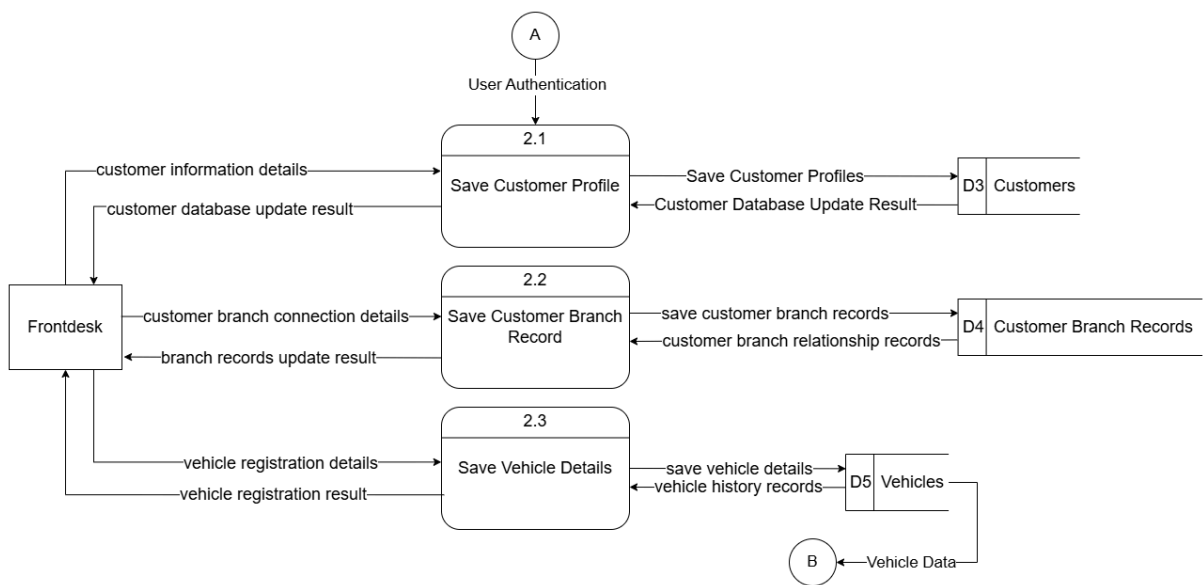
The “Manage Login and User Access” explosion shows how the system handles user registration, authentication, and administration. Users provide registration information, which the system stores to create registered user profiles. When users log in, the system verifies their



credentials and returns login results, granting or denying access based on the checks. The Admin/Owner can update user accounts and view updated user lists, while also managing branch configurations and recording branch setup results. All user data and branch details are stored in the corresponding databases, ensuring accurate record-keeping and secure user authentication while supporting smooth branch administration and system access management.

Figure 6

2.0 Manage Customer and Vehicle Records Explosion

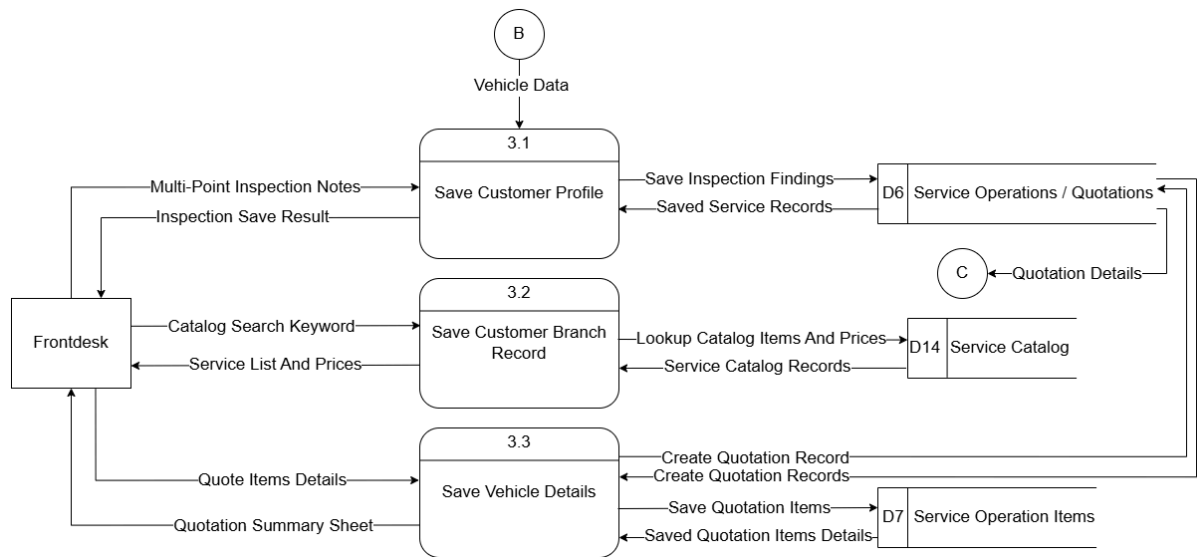


The “Manage Customer and Vehicle Records” explosion illustrates how the system handles customer and vehicle information. Front desk staff enter customer details, which are saved as customer profiles and updated in the customer database. Customer branch connections are recorded to maintain relationships between customers and branches, ensuring accurate branch-level management. Vehicle registration details are also captured, with the system saving both vehicle records and vehicle history for future reference. All data flows through user authentication to ensure secure access, while the system maintains updated records in the corresponding data stores for customers, customer branch relationships, and vehicles, supporting accurate tracking and management of all customer and vehicle information.

Figure 7



3.0Manage Service Operations and Quotations

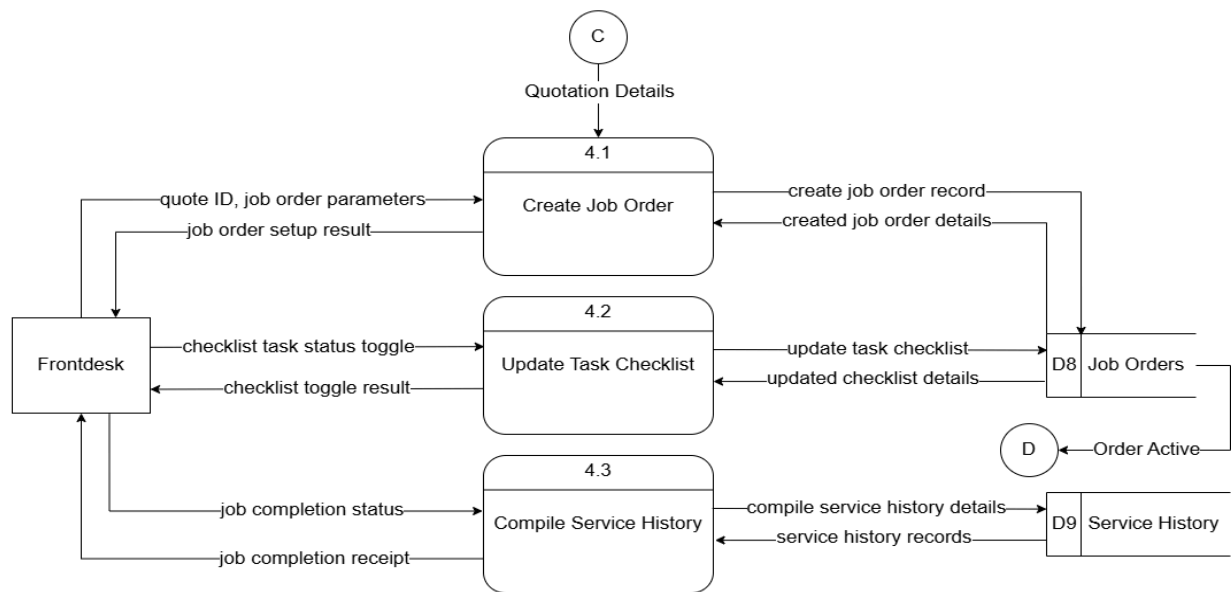


The “Manage Service Operations and Quotations” process shows how the system handles customer service inspections, branch records, and quotation creation. Front Desk staff input multi-point inspection notes and service results for vehicles, which the system saves as inspection findings and service records. Customer branch records are maintained, and the system allows catalog searches to retrieve service items and prices from the service catalog. Quotation details are generated based on selected service items, and the system saves both the quotation records and the individual quotation items. Vehicle data is linked throughout the process, ensuring that inspection results, service items, and quotation information are associated correctly with each customer and vehicle. All data is stored in the appropriate databases, supporting accurate tracking of service operations, quotations, and branch-specific service information for reporting and decision-making.

Figure 8



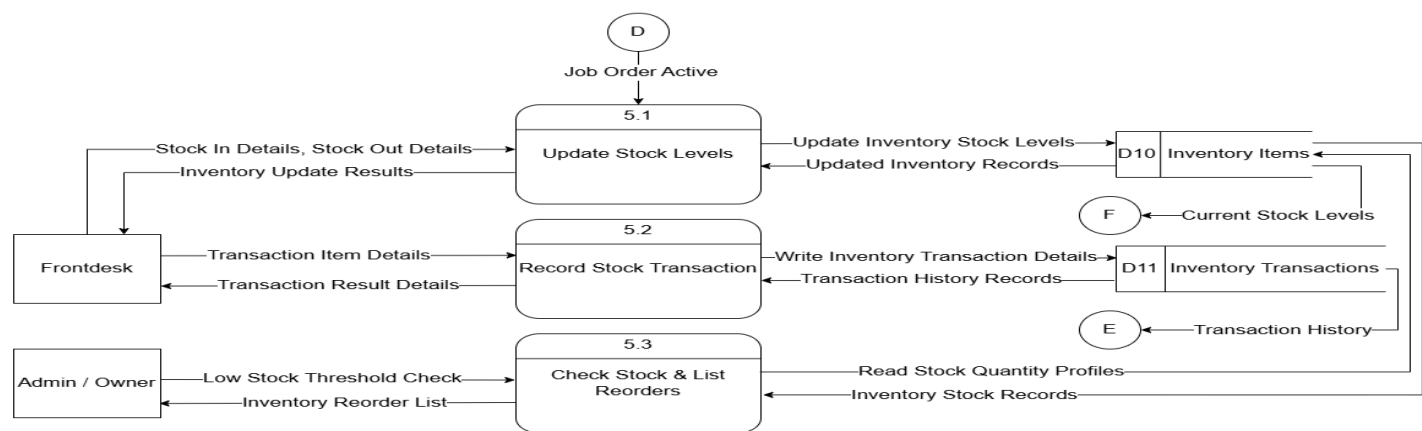
4.0Manage Job Orders and Checklists



The “Manage Job Orders and Checklists” process illustrates how job orders are created, tracked, and recorded in the system. The Front Desk inputs quotation details and job order parameters, which the system uses to generate job order records with all relevant information. The task checklist for each job order is maintained and updated by the Front Desk, allowing the system to track task completion and update the checklist details accordingly. Once a job is completed, the system compiles the service history, ensuring all completed services are recorded and linked to the corresponding job order. This process ensures accurate tracking of job progress, task completion, and service history, providing a clear record for reporting, customer follow-ups, and future service operations.

Figure 9

5.0Manage Inventory and Stock

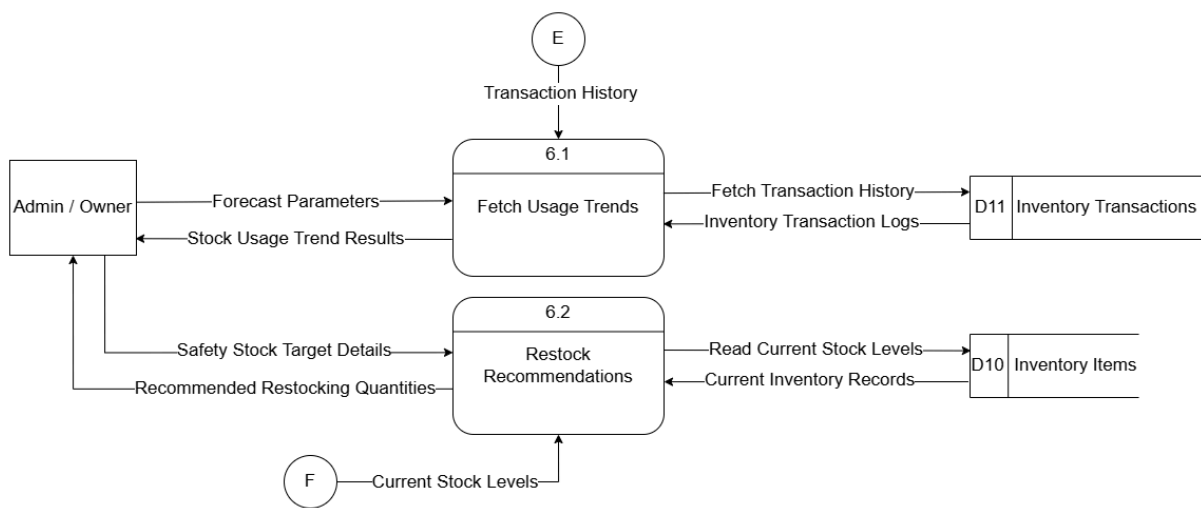




The “Manage Inventory and Stock” process describes how the system tracks, updates, and manages inventory items and stock movements. The Front Desk records stock details, including stock-in and stock-out information, and the system updates the inventory stock levels accordingly. Inventory transactions are recorded with detailed transaction items, which the system stores as transaction history for future reference. Admin/Owner monitors inventory by checking stock quantities, reviewing low-stock alerts, and generating reorder lists. The system ensures that inventory items, stock levels, and transaction histories are accurate and up to date, supporting proper stock control, replenishment planning, and decision-making for ongoing operations.

Figure 10

6.0Manage Forecasting and Decision Support

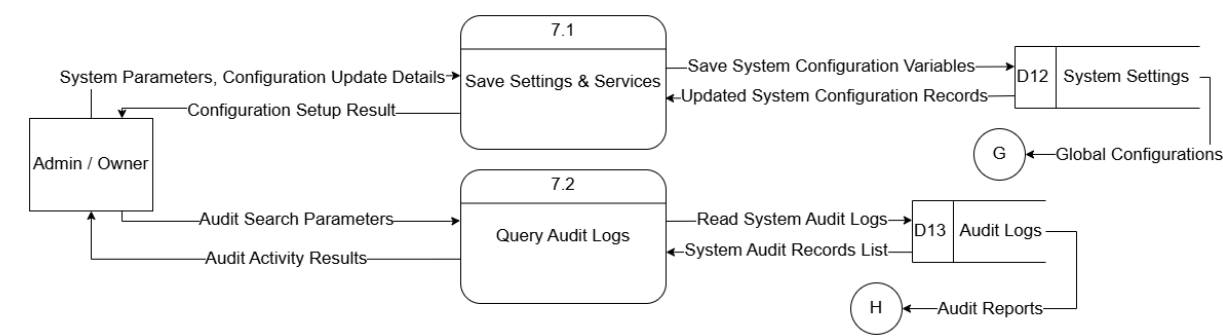


The “Manage Forecasting and Decision Support” process demonstrates how the system analyzes inventory trends and provides restocking recommendations for decision-making. The Admin/Owner inputs forecast parameters, which the system uses to fetch historical transaction data and inventory transaction logs to determine usage trends. Based on these trends and current stock levels, the system calculates recommended restocking quantities and safety stock targets. The Admin/Owner receives the stock usage trend results and restock recommendations to make informed decisions regarding inventory management. This process ensures that inventory levels are maintained efficiently, preventing stockouts and overstock situations, while supporting operational planning and resource allocation across branches.



Figure 11

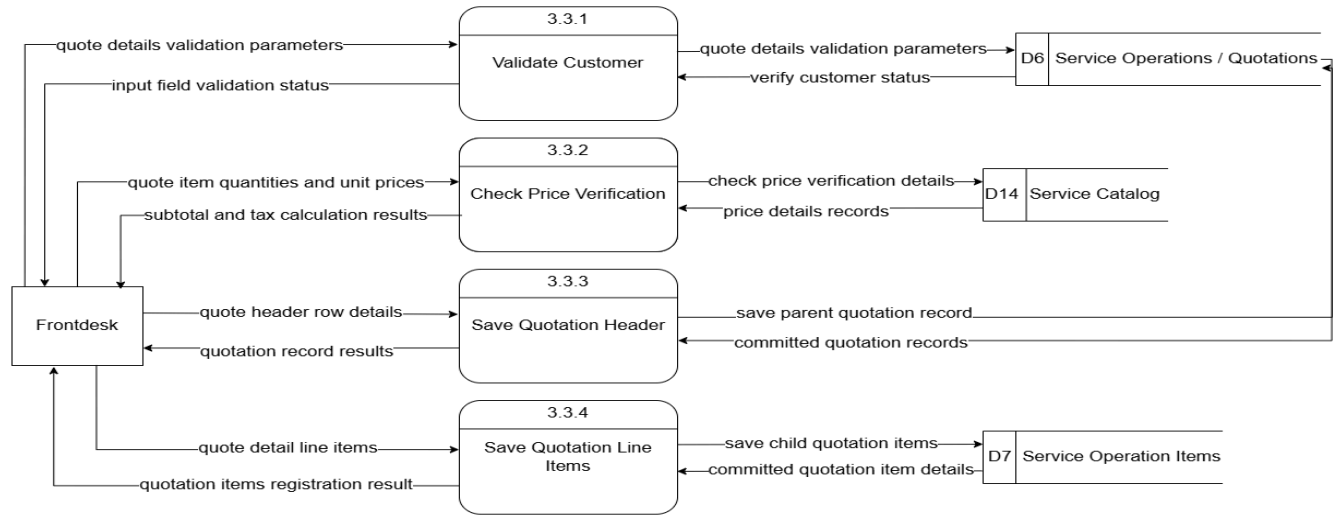
7.0Manage Reports and System Maintenance



The “Manage Reports and System Maintenance” process shows how the Admin/Owner maintains system settings, monitors system activity, and generates reports. System configuration details, parameters, and service updates are entered and saved by the Admin/Owner, with the system validating and storing the configuration variables. The system also allows the Admin/Owner to query audit logs by providing search parameters, and the system retrieves the corresponding audit records and activity results. These records support the generation of audit reports and ensure that system changes and activities are tracked accurately. This process guarantees proper system maintenance, secure tracking of updates and activities, and the availability of reports for monitoring, compliance, and decision-making.

Figure 12

Save Quotation Process Level 3 Explosion

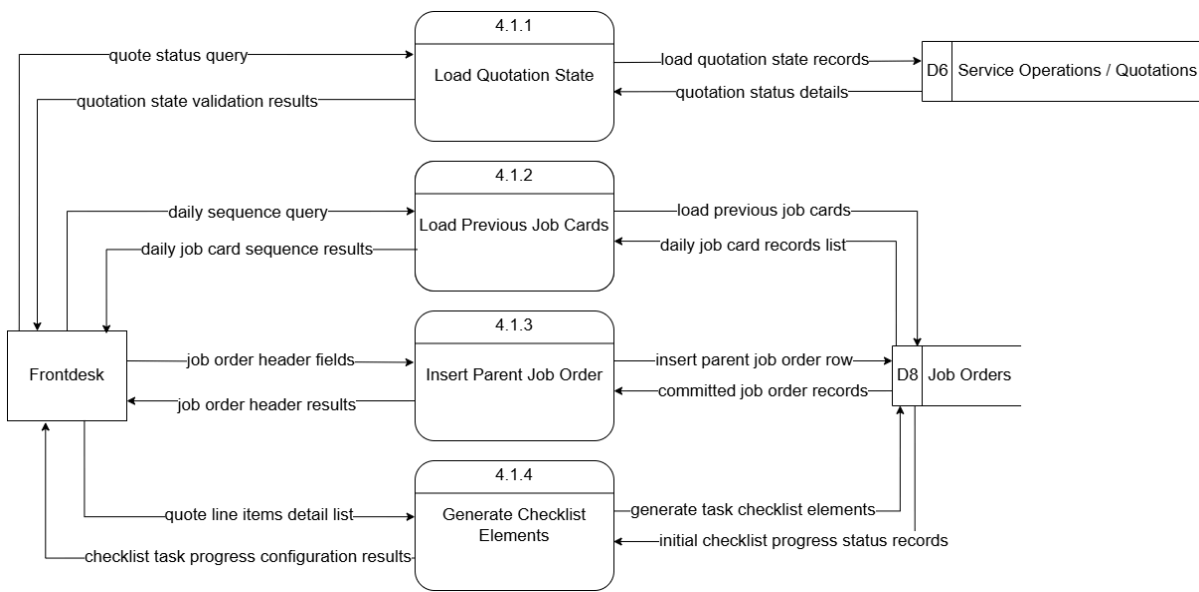




The “Save Quotation Process Level 3 Explosion” process shows how the system handles detailed quotation creation and verification for service operations. The Front Desk inputs customer information, quotation details, and item quantities with unit prices. The system first validates the customer and ensures their status is verified. Next, it checks price verification against the service catalog to calculate subtotals, taxes, and confirm pricing accuracy. The Front Desk then saves the quotation header, creating a parent record for the overall quotation, which the system commits to the service operations database. Finally, the quotation line items are saved as child records, ensuring each service item and its details are accurately registered and linked to the parent quotation. This process ensures accurate customer validation, precise pricing, and proper recording of all quotation details for future reference, approvals, and job order creation.

Figure 13

Convert Quote to Job Order Process Level 3 Explosion



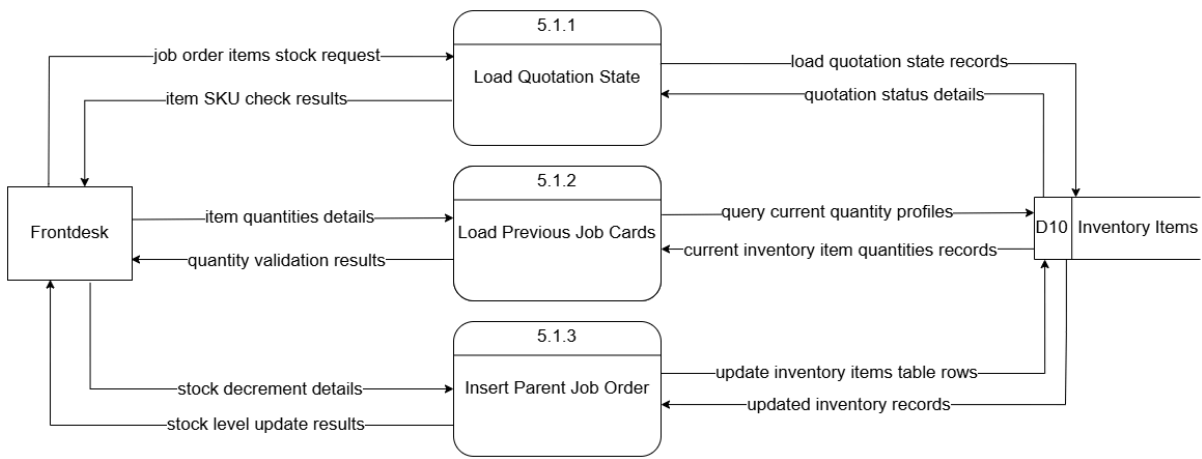
The “Convert Quote to Job Order Process Level 3 Explosion” process illustrates how the system transforms approved quotations into job orders for service execution. The Front Desk begins by loading the quotation state records to verify the status of each quote. Previous job cards are then loaded to maintain sequential job order numbering and avoid duplication. The system allows the Front Desk to insert a parent job order row based on the quotation header, creating a committed record in the job orders database. Next, the system generates checklist elements for each job, using the quotation line items to create tasks and initial progress status records. This process ensures that quotations are accurately converted into actionable job orders,



with all associated tasks and sequences correctly recorded, enabling structured service execution and proper tracking of job progress.

Figure 14

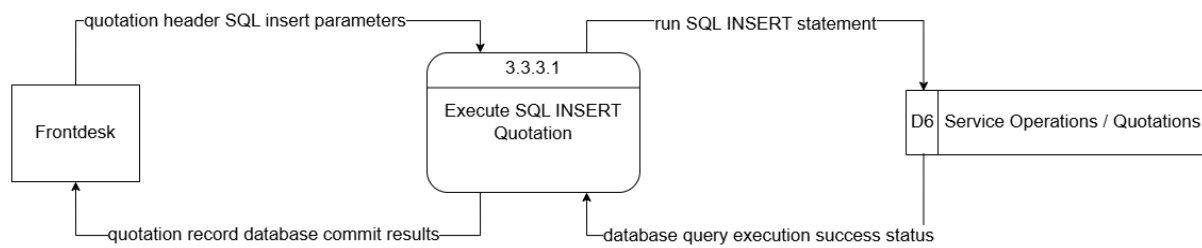
Update Stock Quantities Process Level 3 Explosion



The “Update Stock Quantities Process Level 3 Explosion” process shows how the system manages and updates inventory stock levels based on job order requirements. The Front Desk begins by loading the quotation state records to verify which items are required for job orders. Previous job cards are then loaded to check the current stock quantities and validate the available inventory. The system then updates the inventory by inserting the parent job order and decrementing stock quantities accordingly, ensuring that all inventory item records and table rows are updated. This process guarantees accurate stock management, prevents stock discrepancies, and ensures that job orders are fulfilled with the correct inventory, while maintaining a complete record of updated inventory levels for reporting and decision-making.

Figure 15

Commit Parent Quotation Row Primitive Level 4 Explosion

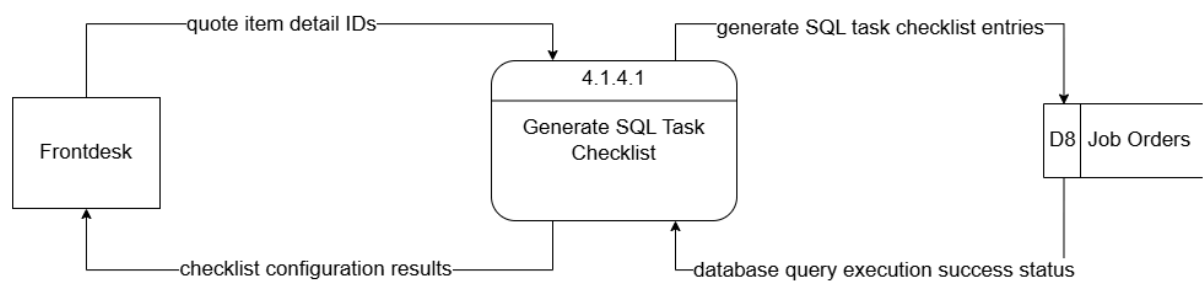




The “Commit Parent Quotation Row Primitive Level 4 Explosion” process illustrates how the system executes the final step of saving a quotation header into the database. The Front Desk provides the SQL insert parameters for the quotation header, which the system uses to run an SQL INSERT statement. Upon successful execution, the database confirms the commit and returns the results to the Front Desk. This process ensures that the parent quotation record is accurately stored in the service operations database, maintaining data integrity and providing a reliable reference for all subsequent quotation and job order operations.

Figure 16

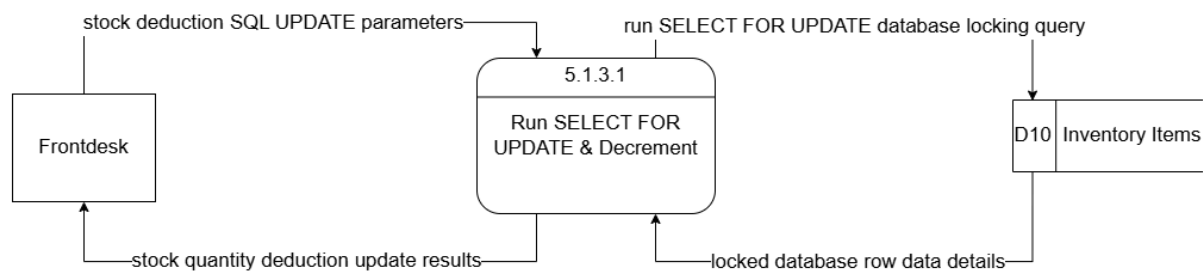
Initialize Progress Checklist Primitive Level 4 Explosion



The “Initialize Progress Checklist Primitive Level 4 Explosion” process shows how the system generates the initial task checklist for a job order based on the quotation items. The Front Desk provides the quote item detail IDs, which the system uses to generate SQL entries for each task in the checklist. Once the SQL statements are executed successfully, the database returns the results to the Front Desk, confirming that the task checklist has been created. This process ensures that all tasks associated with a job order are properly initialized in the system, enabling accurate tracking of job progress and completion status for service operations.

Figure 17

Deduct Stock Quantity Locking Primitive Level 4 Explosion





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The “Deduct Stock Quantity Locking Primitive Level 4 Explosion” process shows how the system ensures accurate inventory management when job orders require stock items. The Front Desk provides the SQL update parameters for the items to be deducted from inventory. The system executes a “SELECT FOR UPDATE” query to lock the relevant database rows, preventing other transactions from modifying the same inventory records simultaneously. Once the stock quantities are decremented successfully, the system returns the results to the Front Desk. This process guarantees that inventory levels are updated safely, avoiding conflicts or discrepancies when multiple job orders or stock operations occur concurrently.



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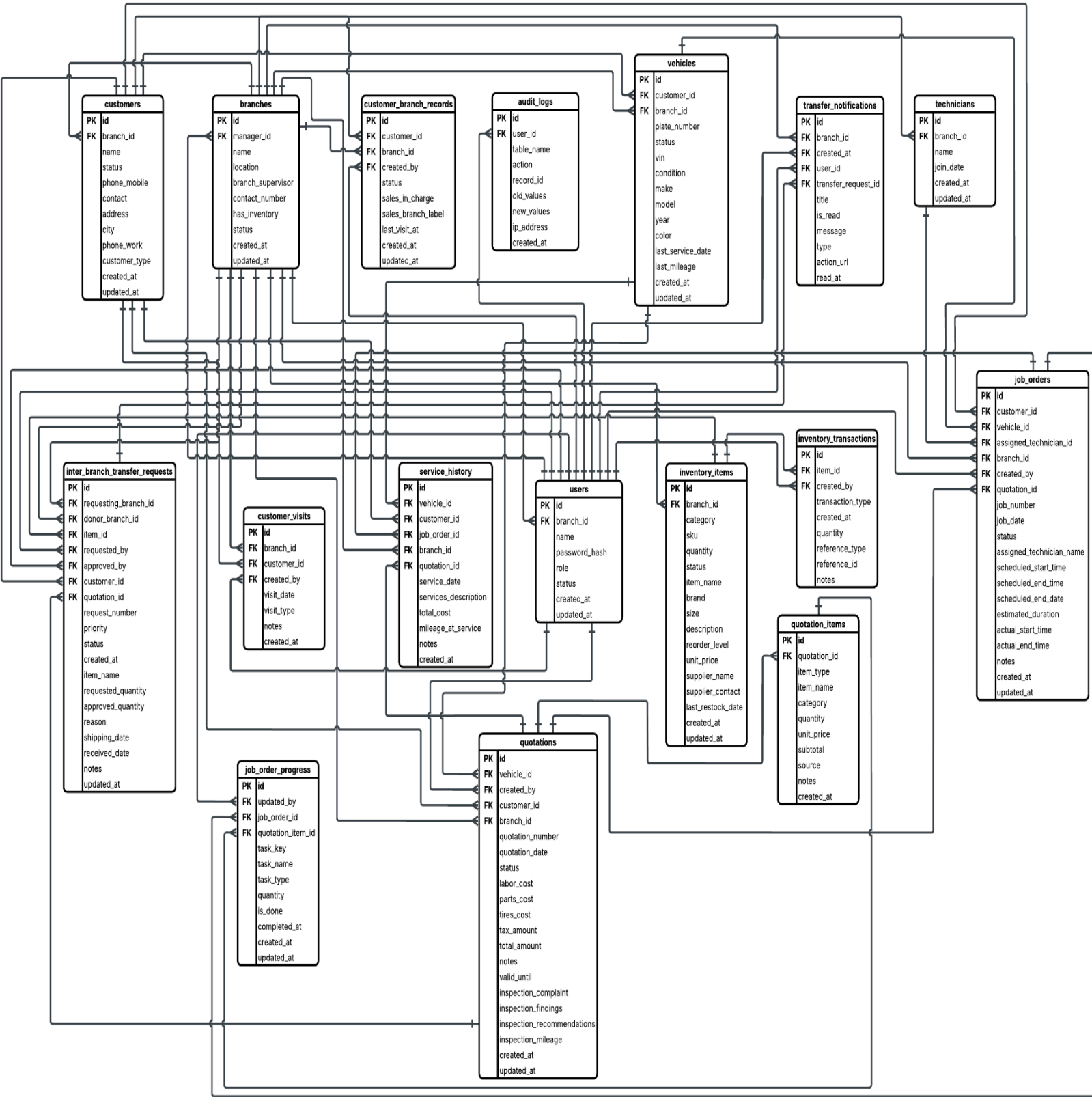
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Figure 18

Entity Relationship Diagram



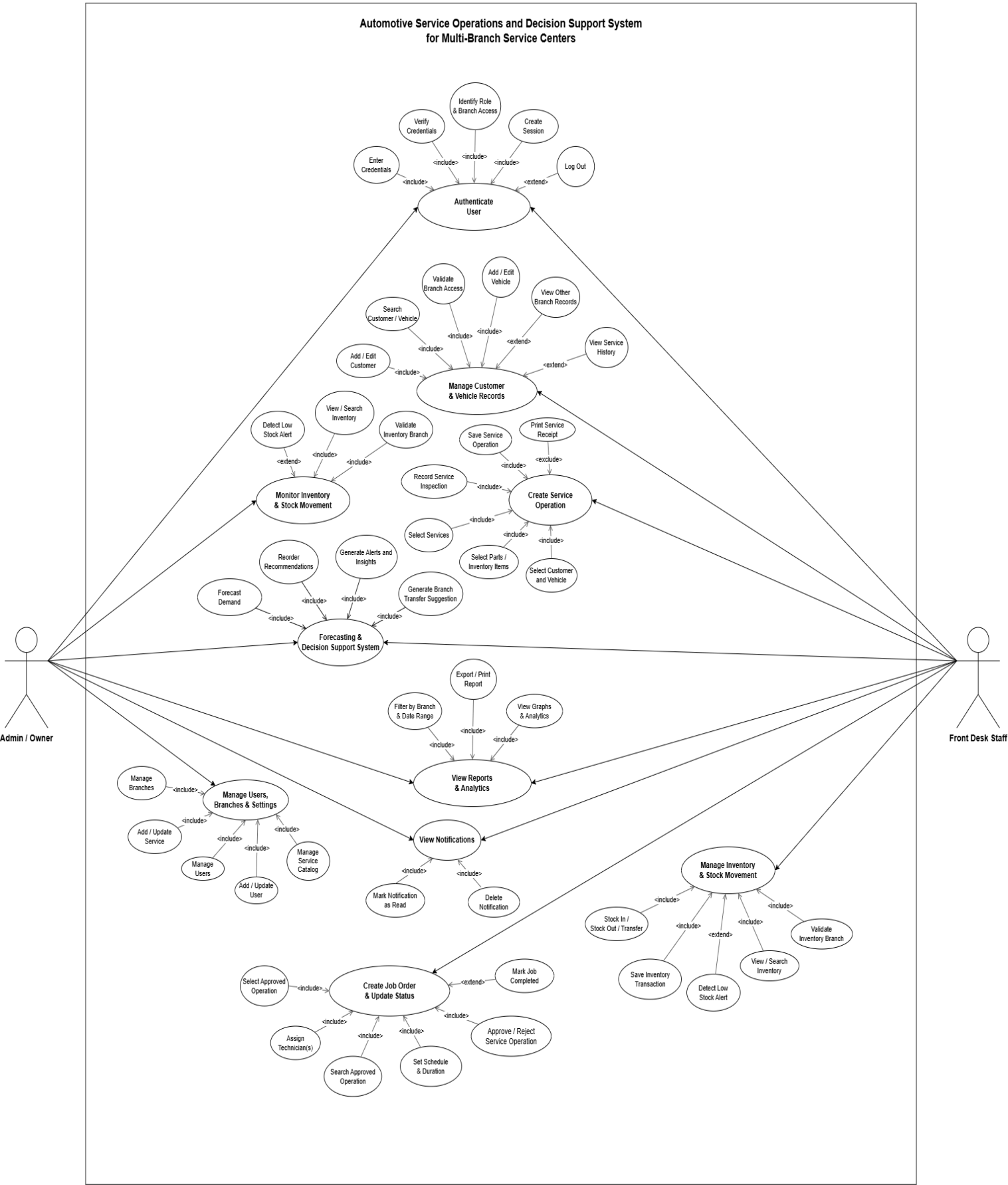


As shown in the previous page, Figure 18 presents the ERD for the proposed Automotive Service Operations and Decision Support System for Multi-Branch Service Centers. The diagram illustrates the overall database structure of the system and the way all major entities are connected to support multi-branch operations. The users table represents the system users and contains information such as login credentials, roles, and branch assignments, enabling authentication and authorization for both Front Desk and Admin / Owner. The branches table stores branch-specific data and is linked to users, customers, vehicles, inventory, and job orders, supporting centralized and branch-specific operations. Customers are recorded in the customers table and are connected to vehicles, service history, visits, and branch-specific records, allowing tracking of customer interactions, service history, and cross-branch activity. Vehicles are linked to customers, job orders, service history, and quotations, enabling management of service scheduling, parts, and maintenance history. Job orders and quotation tables manage service requests, tasks, and cost details, while the quotation_items and job_order_progress tables track individual service items, quantities, and progress. Inventory is managed through inventory_items and inventory_transactions tables, tracking stock levels, movements, and low-stock alerts. Additional tables such as transfer_notifications, audit_logs, system_settings, and service_catalog provide support for notifications, auditing, configuration, and reference data. The ERD shows that the system maintains centralized customer and vehicle records, supports job order processing, inventory tracking, and reporting, and enables decision support for inventory replenishment and service planning. This structure reflects the proposed multi-branch, integrated, and decision-support nature of the system, ensuring efficient operational management, reporting, and forecasting across all branches.



Figure 19

UML Use Case Diagram





As shown in the previous page, Figure 19 presents the Use Case Diagram for the proposed Automotive Service Operations and Decision Support System for Multi-Branch Service Centers. The diagram highlights the major interactions between the system and its primary users: the Front Desk and the Admin / Owner. The Front Desk is responsible for day-to-day operational tasks, including managing customer and vehicle records, processing service operations, generating quotations, creating job orders, updating job progress, and recording service history. Through these functions, the Front Desk can capture customer concerns, log inspection findings, approve and save quotations, schedule service operations, and maintain accurate vehicle service records. Meanwhile, the Admin / Owner oversees higher-level functions such as managing users and branches, configuring system settings, monitoring inventory and stock levels, generating reports and analytics, and reviewing forecasting outputs for decision support. The diagram shows that both actors interact with centralized entities such as users, branches, customers, vehicles, inventory items, job orders, quotations, and service history, ensuring consistent and synchronized data across branches. These interactions facilitate effective multi-branch management, operational tracking, inventory control, reporting, and forecasting, reflecting the system’s integrated, decision-support, and service-focused functionality.

Use Case Scenario

Table 1

Use Case 1: Manage Login and User Access

Use Case Goal: To allow the Admin/Owner and Front Desk to access the system securely, manage user roles, and maintain branch access.

Scenario 1: Successful login and authentication.

Scenario 2: Update of user or branch information.

Actor: Admin/Owner, Front Desk

Pre-condition: Users must have valid credentials and system access rights.

Main Flow:

Step 1: The user opens the Login and User Management module.

Step 2: The system validates credentials.

Step 3: The user is authenticated and redirected to the system dashboard.



Step 4: Admin/Owner can create or update users and branches.

Step 5: The system saves updated users and branch records.

User Action: Logs in, manages users, and updates branch details.

System Response: Validates login and saves updates.

Alternative Flow:

Step 1: Invalid login credentials prompt the user to retry.

Step 2: Duplicate usernames or missing branch data prompt corrections.

Post-condition: Users are authenticated, and user/branch records are updated.

Table 2

Use Case 2: Manage Customer and Vehicle Records

Use Case Goal: To allow the Front Desk to record, update, and retrieve customer and vehicle information, including cross-branch service history.

Scenario 1: Successful creation of a new customer and vehicle profile.

Scenario 2: Successful update of existing customer and vehicle records.

Actor: Front Desk

Pre-condition: The user is authenticated.

Main Flow:

Step 1: Open Customer & Vehicle Records module.

Step 2: Enter or update customer details.

Step 3: Save customer profile.

Step 4: Enter branch-specific customer records.

Step 5: Save branch records.

Step 6: Enter vehicle registration and history details.

Step 7: Save vehicle records.

User Action: Inputs or updates customer and vehicle information.

System Response: Validates and saves the data.



Alternative Flow:

Step 1: Missing fields prompt the user to complete information.

Step 2: Duplicate customer or vehicle records trigger an alert.

Post-condition: Customer and vehicle records are updated and branch relationships maintained.

Table 3

Use Case 3: Manage Service Operations and Quotations

Use Case Goal: To allow the Front Desk to record inspections, create quotations, and manage service operations.

Scenario 1: Successful creation of a service record and quotation.

Scenario 2: Update of existing service operation or quotation.

Actor: Front Desk

Pre-condition: The user is authenticated and the customer and vehicle records exist.

Main Flow:

Step 1: Open Service Operations module.

Step 2: Enter inspection findings and service concerns.

Step 3: Search service catalog for items and prices.

Step 4: Add service items to quotation.

Step 5: Save quotation header and line items.

Step 6: Approve quotation if required.

User Action: Inputs inspection, service items, and quotation details.

System Response: Validates and saves service operation and quotation records.

Alternative Flow:

Step 1: Missing data prompts a warning for completion.

Step 2: Duplicate quotations or service items trigger an alert.

Post-condition: Service operations and quotations are recorded and linked to the customer and vehicle.



Table 4

Use Case 4: Manage Job Orders and Checklists

Use Case Goal: To allow the Front Desk to convert approved quotations into job orders and track task progress.

Scenario 1: Job order successfully created from a quotation.

Scenario 2: Task checklist updated during service execution.

Actor: Front Desk

Pre-condition: The user is authenticated and the quotation is approved.

Main Flow:

- Step 1: Open Job Order module.
- Step 2: Load approved quotation details.
- Step 3: Insert parent job order record.
- Step 4: Generate task checklist from quotation items.
- Step 5: Track task progress and update status.

User Action: Creates job orders and updates tasks.

System Response: Validates, saves, and tracks job order and task data.

Alternative Flow:

- Step 1: Missing task details prompt the user for completion.
- Step 2: Duplicate job order detection triggers a warning.

Post-condition: Job orders and checklists are recorded and ready for service execution.

Table 5

Use Case 5: Manage Inventory and Stock

Use Case Goal: To allow Admin/Owner and Front Desk to manage inventory, track stock movements, and monitor low-stock alerts.



Scenario 1: Stock-in, stock-out, and transfer successfully recorded.

Scenario 2: Low-stock alerts generated for inventory replenishment.

Actor: Admin/Owner, Front Desk

Pre-condition: User is authenticated and inventory items exist.

Main Flow:

- Step 1: Open Inventory module.
- Step 2: Add or update inventory items.
- Step 3: Record stock movements.
- Step 4: Validate transactions.
- Step 5: Generate low-stock alerts and reorder recommendations.

User Action: Inputs inventory updates and stock movements.

System Response: Updates stock levels, validates transactions, and triggers alerts.

Alternative Flow:

- Step 1: Missing item data prompts a request for completion.
- Step 2: Duplicate inventory records trigger a warning.

Post-condition: Inventory levels are updated, and stock alerts are generated.

Table 6

Use Case 6: Forecasting and Decision Support

Use Case Goal: To allow Admin/Owner to generate forecasts, inventory recommendations, and support management decisions.

- Scenario 1:** Generate stock usage forecasts.
- Scenario 2:** Generate branch transfer recommendations.
- Scenario 3:** Generate inventory and service reports.

Actor: Admin/Owner

Pre-condition: User is authenticated and historical data exists.



Main Flow:

- Step 1: Open Forecasting and DSS module.
- Step 2: Retrieve historical stock and service data.
- Step 3: Analyze trends and calculate forecasts.
- Step 4: Generate reorder and branch transfer recommendations.
- Step 5: Create reports and analytics for decision-making.

User Action: Requests forecasts, transfers, or reports.

System Response: Produces recommendations and generates reports.

Alternative Flow:

- Step 1: Missing or incomplete data triggers system alerts.
- Step 2: Data anomalies are flagged for review.

Post-condition: Forecasts, recommendations, and reports are available for Admin/Owner decisions.

Table 7

Use Case 7: Manage System Maintenance and Notifications

Use Case Goal: To allow Admin/Owner to update system settings, manage the service catalog, and maintain audit logs and notifications.

Scenario 1: Update system settings and service catalog items.

Scenario 2: Generate notifications for users.

Scenario 3: View audit logs for system activity.

Actor: Admin/Owner

Pre-condition: User is authenticated and has administrative privileges.

Main Flow:

- Step 1: Open System Maintenance module.
- Step 2: Update settings, service catalog, or user notifications.
- Step 3: System validates and saves updates.
- Step 4: Audit logs are updated automatically.



User Action: Updates system parameters and generates notifications.

System Response: Saves updates, logs activity, and distributes notifications.

Alternative Flow:

Step 1: Missing data prompts the user to complete the required fields.

Step 2: Duplicate service catalog items trigger a warning.

Post-condition: System settings, catalog, notifications, and audit logs are updated and maintained.

Data Dictionary

Data Dictionary provides a detailed description of the system’s data elements, including their standard definitions, meaning, data types, formats, field sizes, and allowable values stored in the database tables. It serves as a guide for understanding the structure, organization, and use of data within the proposed system.

Here’s a continuation of your data dictionary based on the ERD you provided, following the same format as your example (Table 9). I have created a new table number for the continuation.

Table 8

Customer Records

Field Name	Data Type	Data Format	Field Size	Description	Example
customer_id	int	Integer	11	Primary key of the 1001 customer record	
branch_id	int	Integer	11	Foreign key referencing 01 the branch where customer is registered	



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name	varchar	String	100	Full name of the customer	Juan Dela Cruz
phone_mobile	varchar	String	15	Customer mobile contact number	09171234567
phone_work	varchar	String	15	Customer work contact number	8845678
contact	varchar	String	50	Alternative contact	Maria Santos
address	varchar	String	255	Customer home or billing address	123 Main St, City
city	varchar	String	50	Customer city	Manila
customer_type	varchar	String	20	Type of customer (e.g., Individual, Corporate)	Individual
status	varchar	String	20	Status of customer (Active/Inactive)	Active
created_at	datetime	YYYY-MM-DD HH:MM	-	Timestamp when customer record was created	2026-06-02 09:30:00
updated_at	datetime	YYYY-MM-DD HH:MM	-	Timestamp when customer record was last updated	2026-06-02 11:00:00

Table 9





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Vehicles

Field Name	Data Type	Data Format	Field Size	Description	Example
vehicle_id	int	Integer	11	Primary key for the vehicle	2001
customer_id	int	Integer	11	Foreign key referencing customer	1001
branch_id	int	Integer	11	Foreign key referencing branch	01
plate_number	varchar	String	15	Vehicle plate number	ABC1234
status	varchar	String	20	Vehicle status (Active/Inactive)	Active
vin	varchar	String	50	Vehicle Identification Number	1HGCM82633A004352
condition	varchar	String	50	Vehicle condition description	Good
make	varchar	String	50	Vehicle make/brand	Toyota
model	varchar	String	50	Vehicle model	Corolla
year	int	Integer	4	Vehicle manufacturing year	2020
color	varchar	String	20	Vehicle color	White
last_service_date	datetime	YYYY-MM-DD	-	Last serviced date	2026-05-10
last_mileage	int	Numeric	10	Last recorded mileage	25,000
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation timestamp	2026-04-15 10:00





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updated_at

datetime

YYYY-MM-DD HH:MM

-

Last updated

2026-06-02 09:30

timestamp

Table 10

Branches

Field Name	Data Type	Data Format	Field Size	Description	Example
branch_id	int	Integer	11	Primary key for the branch	01
manager_id	int	Integer	11	Foreign key referencing manager	1001
name	varchar	String	100	Branch name	Main Branch
location	varchar	String	255	Branch location	123 Main St, City
branch_supervisor	varchar	String	100	Branch supervisor name	Juan Dela Cruz
contact_number	varchar	String	15	Branch contact number	09171234567
has_inventory	varchar	String	5	Indicates if branch manages inventory (Yes/No)	Yes
status	varchar	String	20	Branch status	Active
created_at	datetime	YYYY-MM-DD HH:MM	-	Branch record creation date	2026-01-15 08:00
updated_at	datetime	YYYY-MM-DD HH:MM	-	Branch record last updated	2026-06-02 10:00

Table 11

Job Orders





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Field Name	Data Type	Data Format	Field Size	Description	Example
job_order_id	int	Integer	11	Primary key of job order	3001
customer_id	int	Integer	11	Foreign key referencing customer	1001
vehicle_id	int	Integer	11	Foreign key referencing vehicle	2001
assigned_technician_id	int	Integer	11	Technician assigned	4001
branch_id	int	Integer	11	Foreign key referencing branch	01
created_by	int	Integer	11	User who created the job order	1001
quotation_id	int	Integer	11	Related quotation record	5001
job_number	varchar	String	20	Job order number	JO-2026-01
job_date	datetime	YYYY-MM-DD HH:MM	-	Job order creation date	2026-06-02 09:00
status	varchar	String	20	Job order status	Active
scheduled_start_time	datetime	YYYY-MM-DD HH:MM	-	Scheduled start	2026-06-03 08:00
scheduled_end_time	datetime	YYYY-MM-DD HH:MM	-	Scheduled end	2026-06-03 12:00
estimated_duration	varchar	String	10	Estimated job duration	4h
actual_start_time	datetime	YYYY-MM-DD HH:MM	-	Actual start time	2026-06-03 08:05
actual_end_time	datetime	YYYY-MM-DD HH:MM	-	Actual end time	2026-06-03 12:10
notes	text	Text	-	Notes regarding the job	Oil change completed





Table 12

Quotations

Field Name	Data Type	Data Format	Field Size	Description	Example
quotation_id	int	Integer	11	Primary key for quotation	5001
vehicle_id	int	Integer	11	Foreign key referencing vehicle	2001
created_by	int	Integer	11	User who created quotation	1001
customer_id	int	Integer	11	Customer related to the quotation	1001
branch_id	int	Integer	11	Branch related to quotation	01
quotation_number	varchar	String	20	Quotation identifier	Q-2026-01
quotation_date	datetime	YYYY-MM-DD HH:MM	-	Quotation creation date	2026-06-02 09:00
status	varchar	String	20	Quotation status	Approved
labor_cost	decimal	Numeric	10,2	Labor cost in quotation	1500.00
parts_cost	decimal	Numeric	10,2	Parts cost	2500.00
tires_cost	decimal	Numeric	10,2	Tires cost	0.00
tax_amount	decimal	Numeric	10,2	Tax included in quotation	400.00
total_amount	decimal	Numeric	10,2	Total quotation cost	4400.00



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notes	text	Text	-	Additional notes	Includes oil and filter change
valid_until	datetime	YYYY-MM-DD	-	Expiry date of quotation	2026-06-09
inspection_complaint	text	Text	-	Complaint reported in inspection	Engine noise
inspection_findings	text	Text	-	Findings from inspection	Oil leak detected
inspection_recommendations	text	Text	-	Suggested repairs	Replace gasket
inspection_mileage	int	Numeric	10	Vehicle mileage at inspection	25000
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-06-02 09:00
updated_at	datetime	YYYY-MM-DD HH:MM	-	Record last updated	2026-06-02 10:00

Table 13

Quotation Items

Field Name	Data Type	Data Format	Field Size	Description	Example
quotation_item_id	int	Integer	11	Primary key for quotation item	7001
quotation_id	int	Integer	11	Foreign key referencing quotation	5001
item_type	varchar	String	20	Type of item (Part/Labor/Service)	Part
item_name	varchar	String	100	Name of the item	Oil Filter
category	varchar	String	50	Category of item	Engine
quantity	int	Numeric	10	Number of items	1





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unit_price	decimal	Numeric	10,2	Price per unit	500.00
subtotal	decimal	Numeric	10,2	Total price for item	500.00
source	varchar	String	50	Source of item	Local Supplier
notes	text	Text	-	Additional remarks	N/A
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-06-02 09:05

Table 14

Inventory Items

Field Name	Data Type	Data Format	Field Size	Description	Example
item_id	int	Integer	11	Primary key for inventory item	9001
branch_id	int	Integer	11	Branch that holds the item	01
category	varchar	String	50	Item category	Oil & Lubricants
sku	varchar	String	50	Stock Keeping Unit	OIL-001
quantity	int	Numeric	10	Quantity in stock	20
status	varchar	String	20	Status of item (Available/Out of Stock)	Available





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item_name	varchar	String	100	Name of item	Engine Oil SAE 5W-30	
brand	varchar	String	50	Brand of item	Mobil	
size	varchar	String	50	Item size or volume	1L	
description	text	Text	-	Item description	Synthetic engine oil	
reorder_level	int	Numeric	10	Minimum threshold	stock	5
unit_price	decimal	Numeric	10,2	Price per unit	500.00	
supplier_name	varchar	String	100	Supplier	OilCo	
supplier_contact	varchar	String	20	Supplier number	contact	09171234567
last_restock_date	datetime	YYYY-MM-DD	-	Last restock date	2026-05-10	
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-01-15 08:00	
updated_at	datetime	YYYY-MM-DD HH:MM	-	Last updated date	2026-06-02 09:30	





Table 15

Inventory Transactions

Field Name	Data Type	Data Format	Field Size	Description	Example
transaction_id	int	Integer	11	Primary key for inventory transaction	10001
item_id	int	Integer	11	Foreign key referencing inventory item	9001
created_by	int	Integer	11	User who performed the transaction	1001
transaction_type	varchar	String	20	Type of transaction (Stock In, Stock Out, Transfer)	Stock In
created_at	datetime	YYYY-MM-DD HH:MM	-	Date and time of transaction	2026-06-02 10:00
quantity	int	Numeric	10	Number of items involved	5
reference_type	varchar	String	50	Reference document type	Purchase Order
reference_id	int	Integer	11	Reference document ID	3001
notes	text	Text	-	Additional notes	Restocked from supplier



Table 16

Service History

Field Name	Data Type	Data Format	Field Size	Description	Example
service_id	int	Integer	11	Primary key for service record	6001
vehicle_id	int	Integer	11	Vehicle serviced	2001
customer_id	int	Integer	11	Customer receiving the service	1001
job_order_id	int	Integer	11	Job order associated with service	8001
branch_id	int	Integer	11	Branch performing the service	01
quotation_id	int	Integer	11	Quotation reference	5001
service_date	datetime	YYYY-MM-DD HH:MM	-	Date of service	2026-06-02 09:30
services_description	text	Text	-	Description of services performed	Oil change, filter replacement



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total_cost	decimal	Numeric	10,2	Total cost of service	4500.00
mileage_at_service	int	Numeric	10	Vehicle mileage during service	25000
notes	text	Text	-	Additional remarks	Customer requested tire check
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-06-02 10:00

Table 17

Customer Branch Records

Field Name	Data Type	Data Format	Field Size	Description	Example
record_id	int	Integer	11	Primary key for customer branch record	4001
customer_id	int	Integer	11	Customer reference	1001
branch_id	int	Integer	11	Branch reference	01
created_by	int	Integer	11	User who created record	1001





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status	varchar	String	20	Record status	Active
sales_in_charge	varchar	String	50	Sales person handling account	Juan Dela Cruz
sales_branch_label	varchar	String	50	Label for sales branch	Talisay Branch
last_visit_at	datetime	YYYY-MM-DD HH:MM	-	Last visit of customer	2026-05-28 10:00
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-01-15 08:00
updated_at	datetime	YYYY-MM-DD HH:MM	-	Last updated date	2026-06-02 09:00

Table 18

Customer Visits

Field Name	Data Type	Data Format	Field Size	Description	Example
visit_id	int	Integer	11	Primary key for customer visit	5001
branch_id	int	Integer	11	Branch visited	01
customer_id	int	Integer	11	Customer reference	1001





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created_by	int	Integer	11	User who logged visit	1001
visit_date	datetime	YYYY-MM-DD HH:MM	-	Date of visit	2026-06-02 09:00
visit_type	varchar	String	20	Type of visit	Service
notes	text	Text	-	Additional remarks	Checked engine noise
created_at	datetime	YYYY-MM-DD HH:MM	-	Record creation date	2026-06-02 09:00

Table 19
Technicians

Field Name	Data Type	Data Format	Field Size	Description	Example
technician_id	int	Integer	11	Primary key for technician	7001
branch_id	int	Integer	11	Branch assigned	01
name	varchar	String	100	Technician name	Pedro Reyes
join_date	datetime	YYYY-MM-DD	-	Date technician joined	2025-01-15





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created_at	datetime	YYYY-MM-DD HH:MM	-	Record date	creation	2026-01-15 08:00
updated_at	datetime	YYYY-MM-DD HH:MM	-	Last updated date		2026-06-02 09:00

Table 20

Transfer Notifications

Field Name	Data Type	Data Format	Field Size	Description	Example
notification_id	int	Integer	11	Primary key for transfer notification	8001
branch_id	int	Integer	11	Branch receiving notification	01
created_at	datetime	YYYY-MM-DD HH:MM	-	Notification creation date	2026-06-02 10:00
user_id	int	Integer	11	User who created notification	1001
transfer_request_id	int	Integer	11	Related transfer request	9001
title	varchar	String	100	Notification title	Item Transfer Request





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is_read	boolean	Boolean	-	Whether notification has been read	False
message	text	Text	-	Notification message	Request transfer 5 Oil units
type	varchar	String	20	Notification type	Transfer
action_url	varchar	String	255	URL for action	/transfer/9001
read_at	datetime	YYYY-MM-DD HH:MM	-	When notification was read	-

Table 21

Audit Logs

Field Name	Data Type	Data Format	Field Size	Description	Example
audit_id	int	Integer	11	Primary key for audit log	10001
user_id	int	Integer	11	User performing the action	1001
table_name	varchar	String	50	Table affected	vehicles
action	varchar	String	50	Type of action	UPDATE





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record_id	int	Integer	11	Record affected	2001
old_values	text	Text	-	Previous values	{"status":"Pending"}
new_values	text	Text	-	New values	{"status":"Completed"}
ip_address	varchar	String	20	User address	IP 192.168.1.10
created_at	datetime	YYYY-MM-DD HH:MM	-	Log creation date	2026-06-02 10:15

Hardware and Software Requirements

Hardware and Software Requirements provided detailed minimum requirements needed to support the developed system's functions and executions

Recommended Minimum Hardware Requirements

- **Computer:** A desktop or laptop computer to host and access the system
- **Processor Speed:** 3.00GHz or higher
- **Processor:** Intel Core i3 or equivalent and above
- **RAM:** At least 8 GB of RAM for smooth system operation
- **Storage:** HDD/SSD with at least 256 GB storage
- **Display:** 1366 x 768 screen resolution or higher
- **Network:** Local network or internet connection for browser-based access
- **Printer:** Optional printer for printing reports and records

Recommended Software Requirements

- **Operating System:** Windows 10 or Windows 11
- **Web Browser:** Google Chrome or any updated web browser
- **Database:** MySQL for data management as primary as the database for data management





- **Programming Language:** PHP and JavaScript
- **Frontend Technologies:** HTML, CSS, Bootstrap, and Font Awesome
- **Text Editor/IDE:** Visual Studio Code

3.9 Project Timeline

The Gantt Chart displayed the project’s timeframe, which spanned from February to November 2025. It divided the work into major phases: project preparation and documentation, system planning and design, system development and testing, and final revision and completion. The chart showed when tasks started and ended, their overlaps, and periods of peak activity. This helped the researchers monitor progress and maintain schedule adherence. The PERT/CPM diagram illustrated the sequence of tasks, their durations, planned start and end times, and slack.

It identified the critical route, indicating tasks that could not be delayed without affecting the overall project timeline. Additionally, it highlighted activities with allowable delays, showing areas of schedule flexibility. Based on the project schedule, the early activities included project preparation and brainstorming, review of related literature and studies, title drafting and approval, and the completion of Chapters 1, 2, and 3. These were followed by system design and planning, UI/UX wireframing, database design, prototype development, testing and debugging for the pre-oral defense, post-pre-oral revisions, feature expansion, system integration, final system testing, and the writing of Chapters 4 and 5. The final stage covered paper revision, adviser review, formatting, printing, and binding. Together, the Gantt Chart and PERT/CPM provided a comprehensive view of the project timeline, supporting effective planning, coordination, and timely completion of the Automotive Service Operations and Decision Support System for Multi-Branch Service Centers.

System Evaluation Tools

Black Box Testing

For the proposed Automotive Service Operations and Decision Support System for Multi-Branch Service Centers, the proponents will use Black Box Testing to evaluate whether each module of the system produces the expected output based on the specified requirements. Black-box testing focuses on the external behavior of a system and is based on the analysis of its specifications rather than its internal code structure (International Software Testing Qualifications Board [ISTQB], 2021).



Through structured test cases, the proponents will verify whether the major modules of the system operate correctly, particularly Branch and User Access Management, Customer and Vehicle Management, Service Operations and Job Order Management, Inventory and Stock Management, Reporting and Consolidated Monitoring, and the Forecasting and Decision Support System. During testing, the proponents will check whether the system correctly accepts valid inputs, rejects invalid inputs, and displays the appropriate results based on the required process flow. In particular, Black Box Testing will be used to verify whether the system properly handles login authentication, customer and vehicle record management, cross-branch service history retrieval, quotation generation, job order creation, service status updates, inventory monitoring, and report generation. This approach is appropriate because it evaluates the system from the users' perspective and confirms whether the expected outputs are produced according to the specified requirements (ISTQB, 2021).

In addition, Wang et al. (2022) reported that higher test automation maturity was positively associated with higher product quality and shorter release cycles. This supports the use of repeatable and structured test cases in evaluating the proposed system and ensuring that its functional requirements are properly met.

White Box Testing

The proponents will also use White Box Testing to examine the internal logic, structure, and flow of the proposed system. White-box testing is a test technique based on the internal structure of a component or system, making it appropriate for validating code-level behavior such as conditions, control flow, computations, and validation rules (International Software Testing Qualifications Board [ISTQB], 2021). For the proposed Automotive Service Operations and Decision Support System for Multi-Branch Service Centers, this testing method will be suitable for checking the internal processes of authentication, customer and vehicle record handling, quotation computation, job order creation, inventory monitoring, reporting functions, and forecasting-related logic.

Since the system will be developed using PHP, the proponents may use PHPUnit and Pest PHP as testing tools for White Box Testing. PHPUnit is described by its official site as an industry-leading test automation framework for PHP, while Pest PHP is presented as a PHP testing framework focused on simplicity and readability. These tools are appropriate for writing automated tests that verify whether the internal functions of the system behave correctly and



consistently at the code level. In addition, Wang et al. (2022) found that stronger test automation maturity was associated with higher product quality and shorter release cycles, supporting the use of automated code-level testing in software development.

Through the use of Pest PHP, the proponents will be able to identify logical errors, failed conditions, and inconsistencies in the internal processes of the proposed system. Failed test cases will serve as indicators of defects in the code, allowing the proponents to review and correct the affected functions. After making the necessary corrections, the same set of automated tests will be executed again to verify whether the issues have been resolved and to ensure that no new errors are introduced. This repeated testing process will help improve the reliability, stability, and maintainability of the system. The use of automated unit testing for repeated verification is also consistent with current research emphasizing its role in improving software quality and supporting faster, more dependable release cycles (Wang et al., 2022).

The ISO/IEC 25010:2011 Software Quality Model was used as the evaluation tool for assessing the quality of the developed Automotive Service Operations with Decision Support System for Multi-Branch Service Centers (Highway Tires Management System) because it provides a standardized and comprehensive way of measuring software performance. This model evaluates systems based on eight characteristics such as Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. These characteristics help determine how well a system meets user needs and performs under real operational conditions.

Each characteristic focuses on a specific aspect of software quality. Functional Suitability checks if the system performs its intended tasks correctly. Performance Efficiency examines the speed of the system and how it uses resources. Compatibility measures how well the system operates alongside other platforms or software. Usability focuses on how easy the system was to learn and use. Reliability ensures that the system works consistently without errors. Security protects information and limits access to authorized users. Maintainability refers to how easily the system can be improved or updated. Portability evaluates the system's ability to function on different devices or environments.

In this study, respondents rated the system using a five points Likert scale ranging from Very Low to Very High based on their experience with functions such as customer-vehicle profiling, multi-point service encoding, real-time stock monitoring, predictive inventory forecasting, and automated multi-branch reporting. The ratings were analyzed using the ISO/IEC



25010:2011 criteria. The results showed that the system achieved very high performance across most areas, with compatibility identified as the area that could be further improved to enhance future interoperability with other platforms and systems.

Likert Scale, Range, and Verbal Interpretation for ISO 25010:2011 System Software Usability Evaluations

Responses	Scale	Range	Verbal Interpretation
Strongly Agree	5	4.20 - 5.00	Very High
Agree	4	3.40 - 4.19	High
Undecided	3	2.60 - 3.39	Average
Disagree	2	1.80 - 2.59	Low
Strongly Disagree	1	1.00 - 1.79	Very Low

Respondents of the Study

Respondents	f	%
Beneficiary	8	34.78%
Client	15	65.22%
Total	23	100%

The table above shows the respondents of the study which consisted of two major groups who play distinct roles in the daily operations of the Automotive Service Operations with Decision Support System for Multi-Branch Service Centers (Highway Tires Management System). These respondents are composed of a total of twenty-three (23) individuals divided into Beneficiaries and Clients.

The Beneficiary group consists of eight (8) individuals representing 34.78% of the total respondents, comprising business owners, branch administrators, and supervisors who oversee operations and benefit directly from the system's management, reporting, and forecasting functionalities. They manage user accounts, configure branch settings, approve or reject service operations, oversee inventory levels, generate forecasting analytics, and review system-wide reports.

The Client group consists of fifteen (15) Frontdesk Staff, representing 65.22% of the total respondents. As the direct end-users of the system, they interact with the client-side interface on a daily basis. They are responsible for encoding day-to-day transaction records, registering



customers, encoding vehicle details, creating service operations and quotations, managing job orders, and processing inventory stock in and stock out transactions.

Their combined perspectives help determine how well the system addresses workflow efficiency, ease of use, decision-making support, and overall satisfaction. This sample of twenty-three (23) respondents represents the key stakeholders, decision-makers, and hands-on users who directly operate the system. This ensures that the feedback collected is grounded in actual system usage and that the system that was created is user-centered, works well in practice, and meets the operational demands of the Highway Tires Management System.

Ethical Consideration

As researchers were developing an Automotive Service Operations with Decision Support System for Multi-Branch Service Centers (Highway Tires Management System), we recognized the importance of ethical considerations in this study. Our commitment to ethical conduct was integral to safeguarding the rights, privacy, and well-being of the individuals involved. Before conducting the evaluation with respondents, we obtained informed consent. Participants were fully informed about the purpose, nature, and potential uses of the study. All information revealed during the evaluation was kept confidential. We ensured that any private information shared was secure. We promised to use the information solely to develop and improve the management system for Highway Tires. The identities of all respondents were kept anonymous throughout the study, and participation was entirely voluntary. No respondent was coerced or pressured into providing feedback, and all collected data was used exclusively for academic and system improvement purposes.